

$$\begin{aligned}
&\boldsymbol{r}_{1\text{CM}\parallel} \text{ and } \boldsymbol{r}_{2\text{CM}\parallel} \\
&\boldsymbol{r}_{1\text{CM}} \text{ and } \boldsymbol{r}_{2\text{CM}} \\
&\boldsymbol{R}_{12} \boldsymbol{R}_{1'2'} \\
&1' \ 2' \ 3' \\
&\boldsymbol{r}_{1_{1-2}} \text{ and } \boldsymbol{r}_{2_{1-2}} \\
&1\text{-}2 \ (1'\text{-}2') \text{ center of mass} \\
&\boldsymbol{r}_{1_{1-2}\perp} \ \boldsymbol{r}_{2_{1-2}\perp} \\
&\boldsymbol{r}_{1_{1-2}\parallel} \ \boldsymbol{r}_{2_{1-2}\parallel} \\
&\boldsymbol{r}_{1_{1-2}\perp_{1-2-3}\parallel}
\end{aligned}$$

$$\frac{V_{i_\alpha}}{\sum V_{i_\alpha}} \tag{1}$$

$$\text{PED}_{i_\alpha} = \frac{V_{i_\alpha}}{\sum V_{i_\alpha}} = \frac{k_i A_{i_\alpha}^2}{\sum k_i A_{i_\alpha}^2} \tag{2}$$

$$\frac{V_{i_{\alpha+1}}}{\sum V_{i_{\alpha+1}}} \tag{3}$$

$$m_1 \boldsymbol{r}_{1\text{CM}}^2 + m_2 \boldsymbol{r}_{2\text{CM}}^2 = \frac{m_1 m_2}{m_1 + m_2} (\boldsymbol{r}_1 - \boldsymbol{r}_2)^2$$

(4)

a b